

# SMARTX: Porewater Sulfide

May 2025 Samples

2026-05-27

## Code Set up

### run information

```
##Before or After Reruns (YES or NO)
After_Reruns = "YES"
#If NO will not write final data file
#And Will Write a Rerun File

###things that need to be changed
sample_month = "May"
sample_year = "2026"
Run_Date = as.Date("05/19/2026", format = "%m/%d/%Y")
Run_by = "Melanie Giessner" #Instrument user
Script_run_by = "Melanie Giessner" #Code user

Run_notes="" #any notes from run

#Import Plate Data

plates<- c("Plate1","Plate2","Plate3","Plate4","Plate5","Plate6","Plate7", "Plate8", "Plate9")

folder_path_rawdata <- paste0("Raw Data")
Raw_plates <- list.files(path = folder_path_rawdata, full.names = TRUE,pattern = "Plate")
head(Raw_plates)

## [1] "Raw Data/20260519_SMARTX_H2S_Plate1.csv"
## [2] "Raw Data/20260519_SMARTX_H2S_Plate2.csv"
## [3] "Raw Data/20260520_SMARTX_H2S_Plate3.csv"
## [4] "Raw Data/20260520_SMARTX_H2S_Plate4.csv"
## [5] "Raw Data/20260520_SMARTX_H2S_Plate5.csv"
## [6] "Raw Data/20260520_SMARTX_H2S_Plate6.csv"

#Import Run Date for Each Plate for Std curve qaqc
Run_dates <- tibble(
  Plate = plates,
  Run_Date = c("05/19/26","05/19/26","05/20/26","05/20/26","05/20/26","05/20/26","05/21/26","05/21/26",
)

Run_dates <- Run_dates %>% mutate(Run_Date = as.Date(Run_Date, format = "%m/%d/%y"))

# Define the file path for QAQC log file - NO Need to change just check year
```

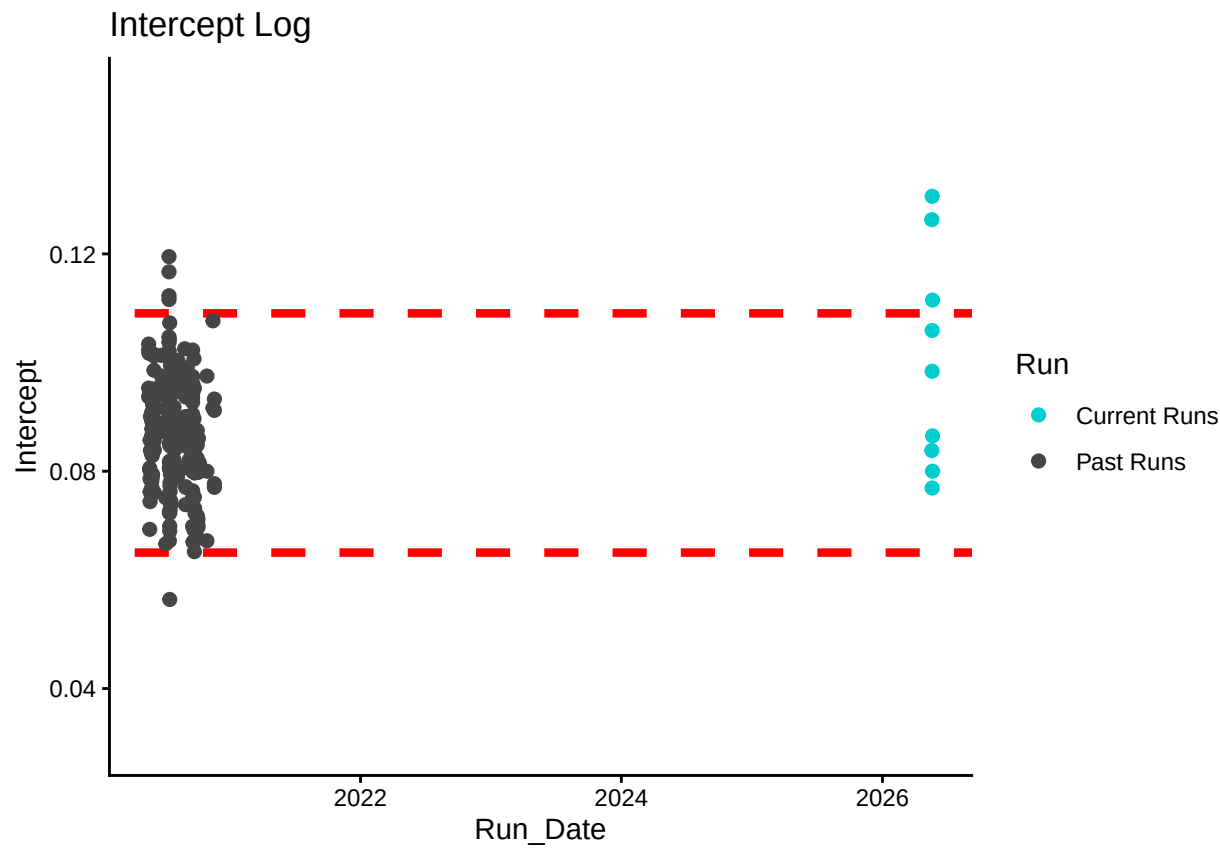
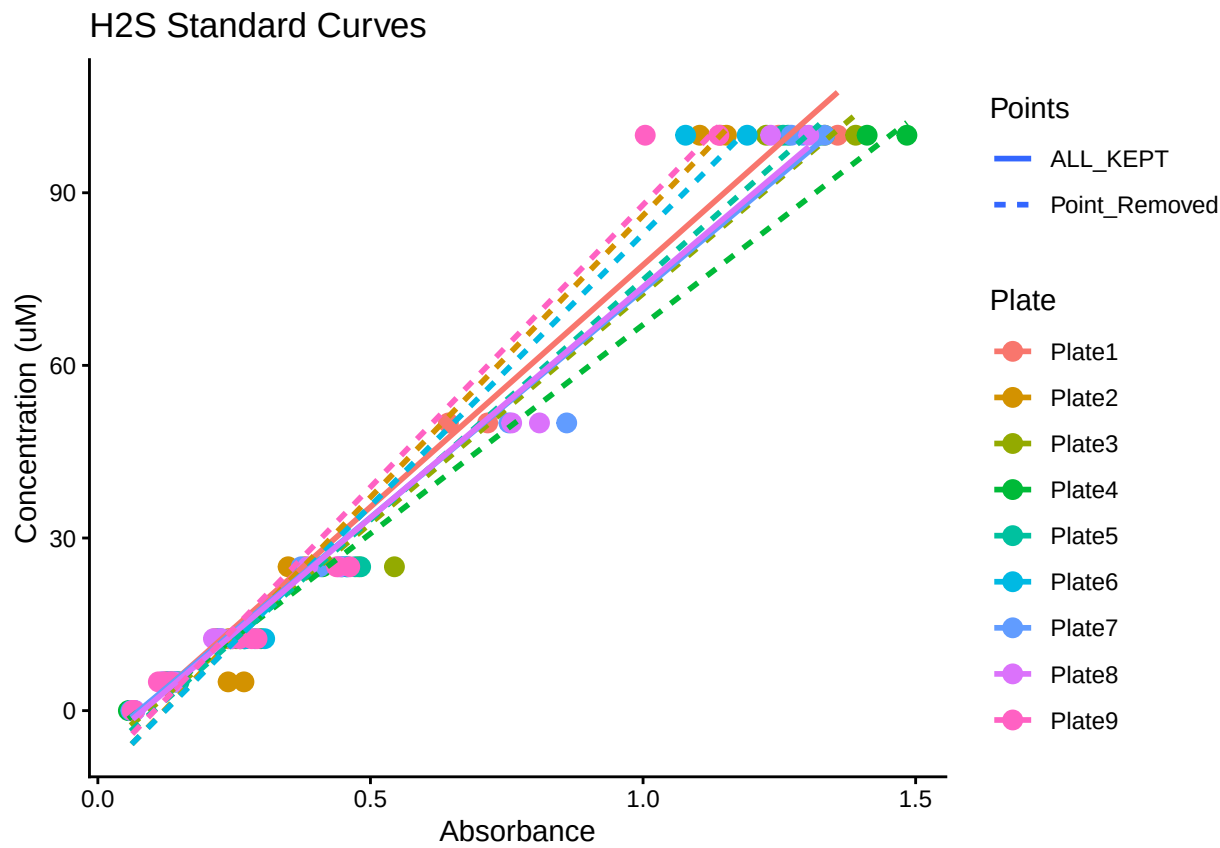
```
log_path <- "Processed Data/SMARTX_H2S_QAQC_2026.csv"

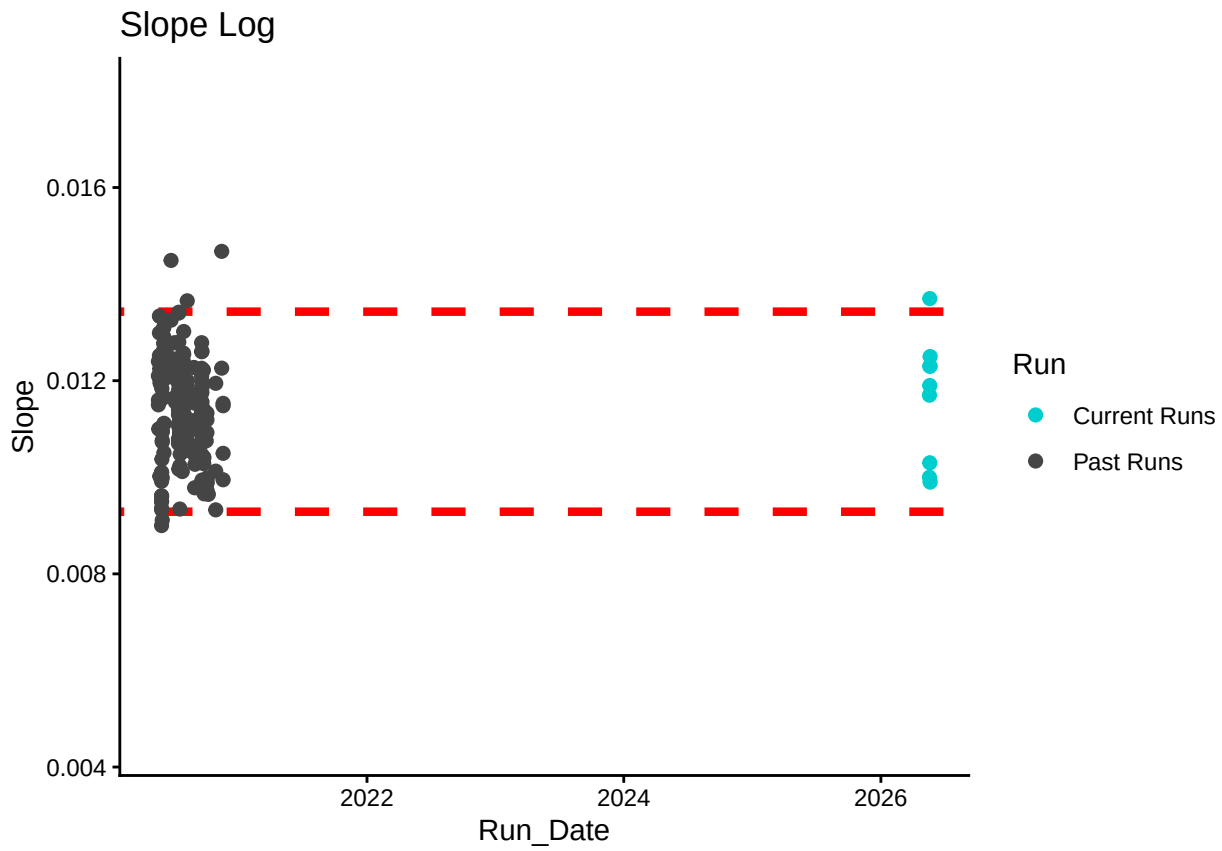
# Define final path for data be sure to change year and month and run number
final_path <- "Processed Data/GCReW_SMARTX_Porewater_H2S_202605.csv"
rerun_path <- "Processed Data/GCReW_SMARTX_Porewater_H2S_202605_reruns.csv"

##Read in metadata and create similar sample IDs for matching to samples
##Read in data
##Remove High CV Stds
##Check R2 & Remove Standards if Necessary

## Joining with 'by = join_by(Plate, IDs)'
```

Plot Standard Curves



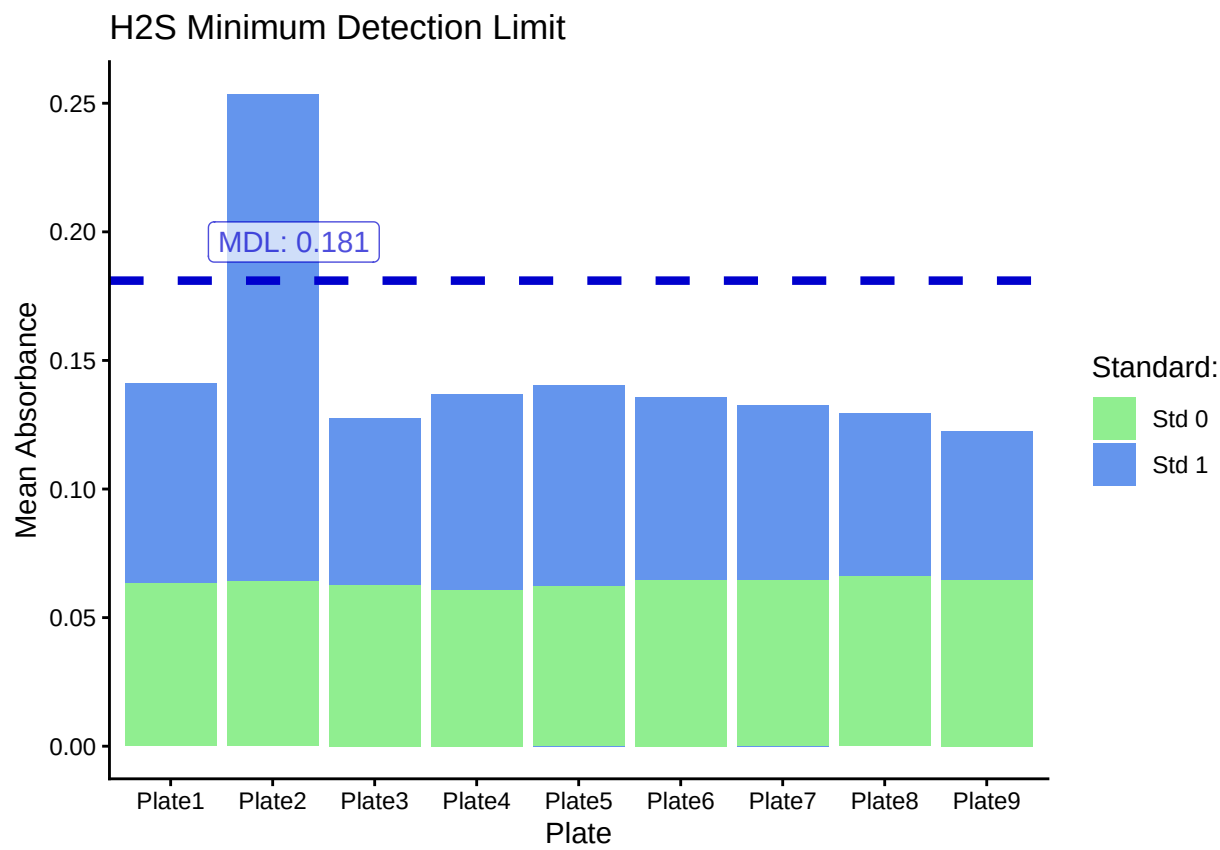


### Return Flags For Standard Curve

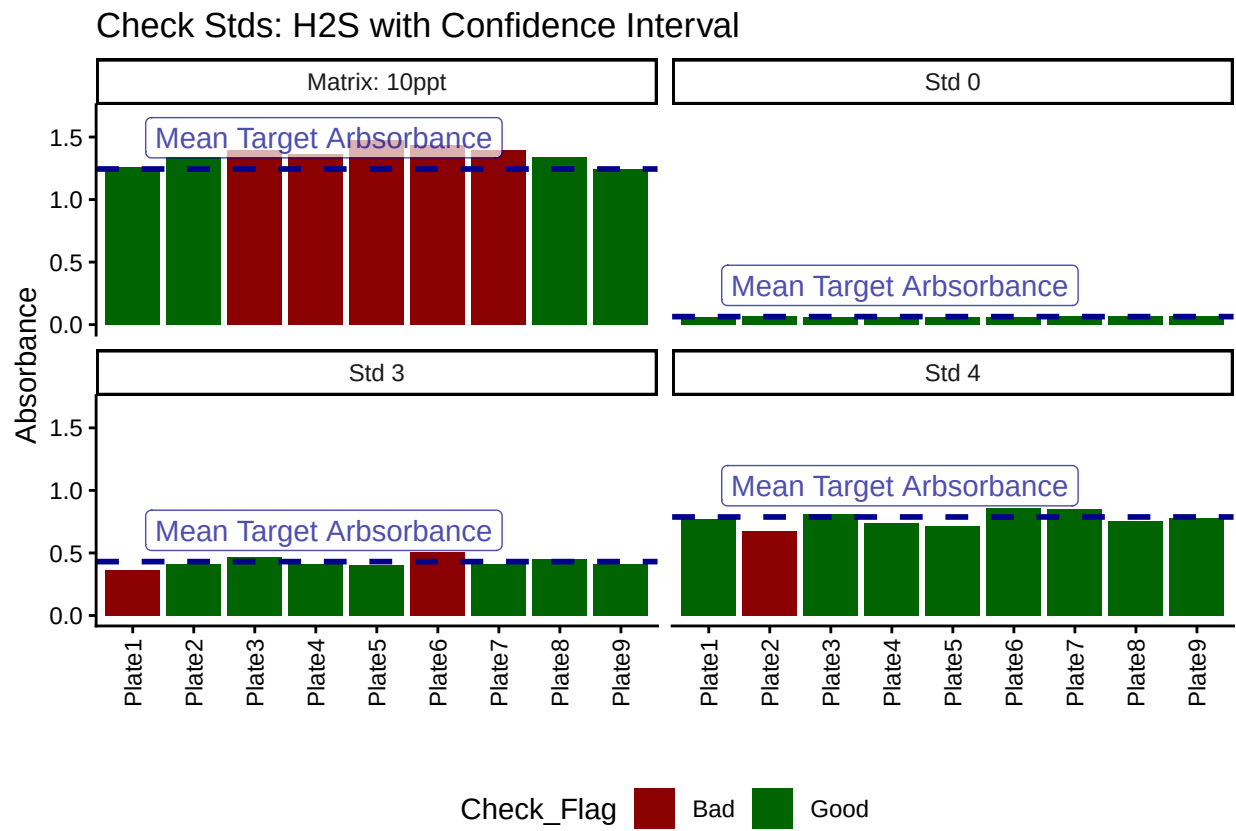
| Plate  | R2_Flag | CV_Flag |
|--------|---------|---------|
| Plate1 | R2_Good | CV_Good |
| Plate2 | R2_Bad  | CV_Good |
| Plate3 | R2_Bad  | CV_Good |
| Plate4 | R2_Good | CV_Good |
| Plate5 | R2_Good | CV_Good |
| Plate6 | R2_Bad  | CV_Good |
| Plate7 | R2_Good | CV_Good |
| Plate8 | R2_Good | CV_Good |
| Plate9 | R2_Bad  | CV_Good |

| Plate  | Curve_RMVD |
|--------|------------|
| Plate1 | KEPT       |
| Plate2 | REMOVED    |
| Plate3 | REMOVED    |
| Plate4 | REMOVED    |
| Plate5 | KEPT       |
| Plate6 | REMOVED    |
| Plate7 | KEPT       |
| Plate8 | KEPT       |
| Plate9 | REMOVED    |

Method Detection Limit



Compare Check Standards to Standards



Display Any Check Flags

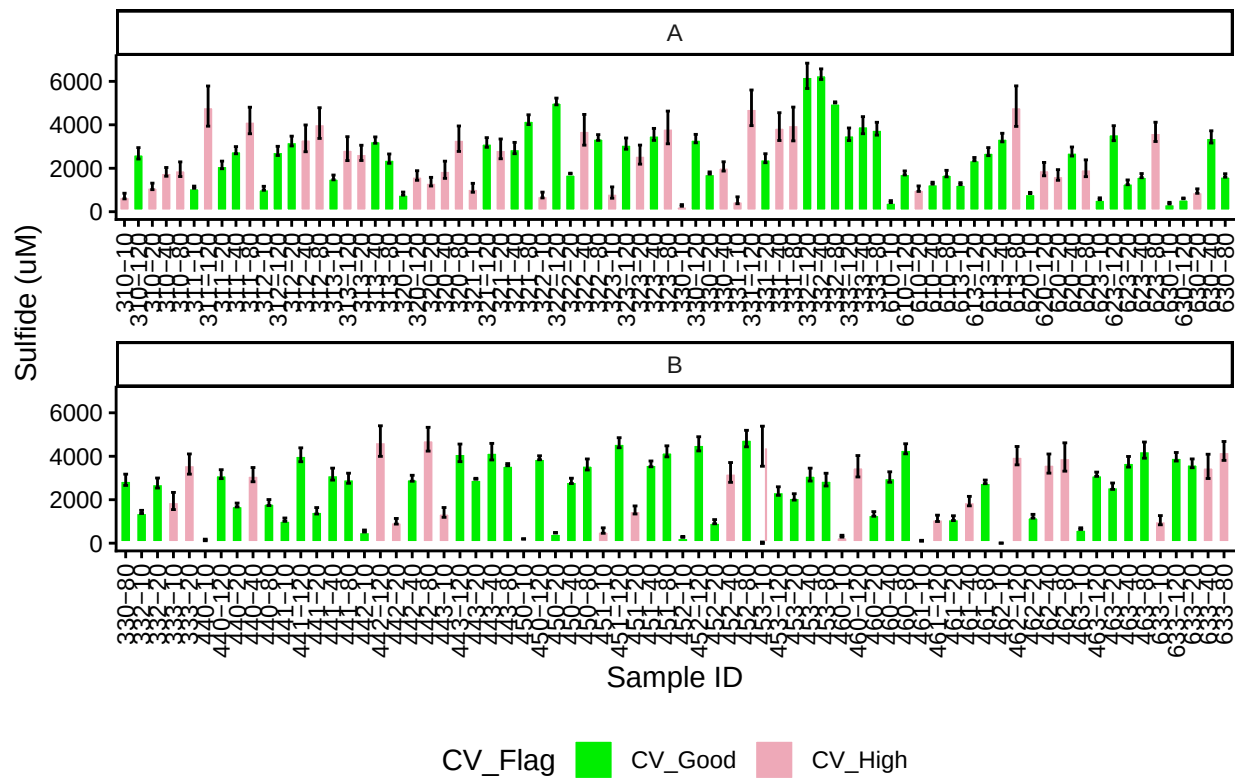
| Plate  | CHKs_Flag   |
|--------|-------------|
| Plate1 | Std3Chk_Bad |
| Plate2 | Std4Chk_Bad |
| Plate3 | Std5Chk_Bad |
| Plate4 | Std5Chk_Bad |
| Plate5 | Std5Chk_Bad |
| Plate6 | Std3Chk_Bad |
| Plate6 | Std5Chk_Bad |
| Plate7 | Std5Chk_Bad |

| Plate | CHK_CV_Flags |
|-------|--------------|
|-------|--------------|

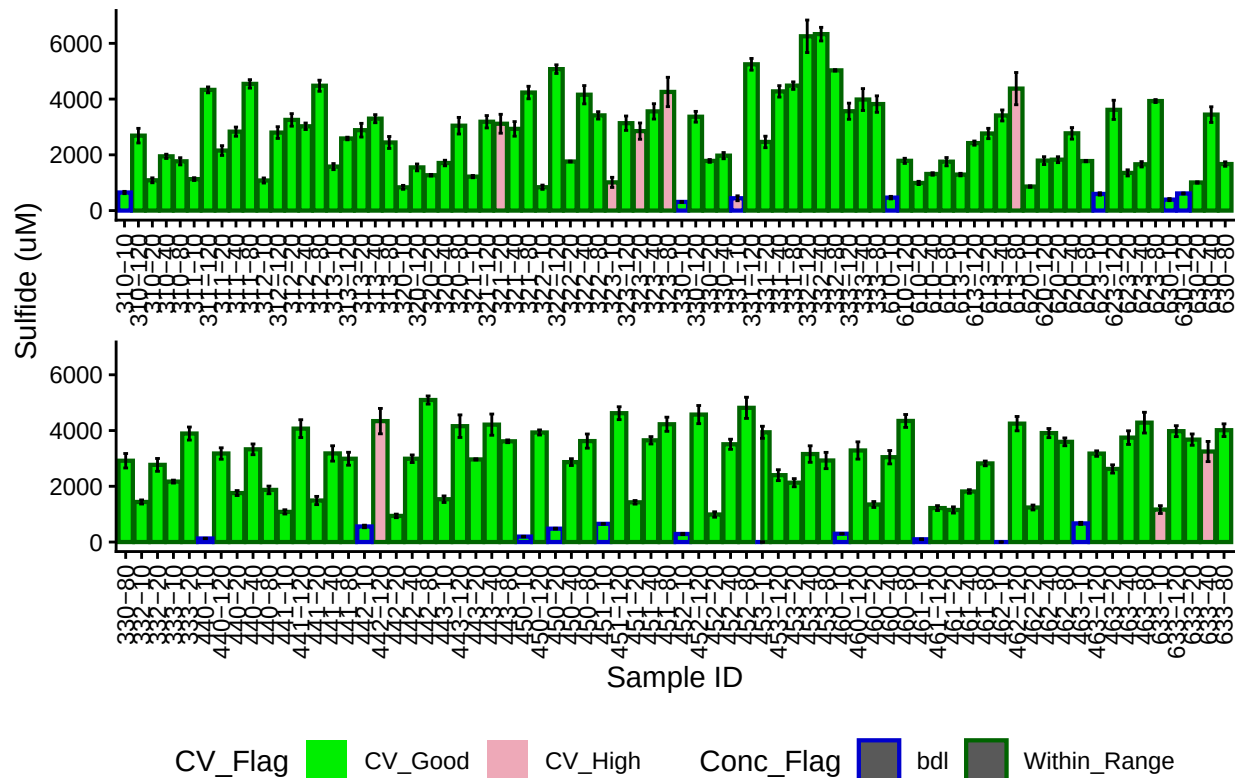
##Calculate Sulfide Concentrations & Add Concentration Flags  
##Remove High CV Samples and Average

Plot Samples

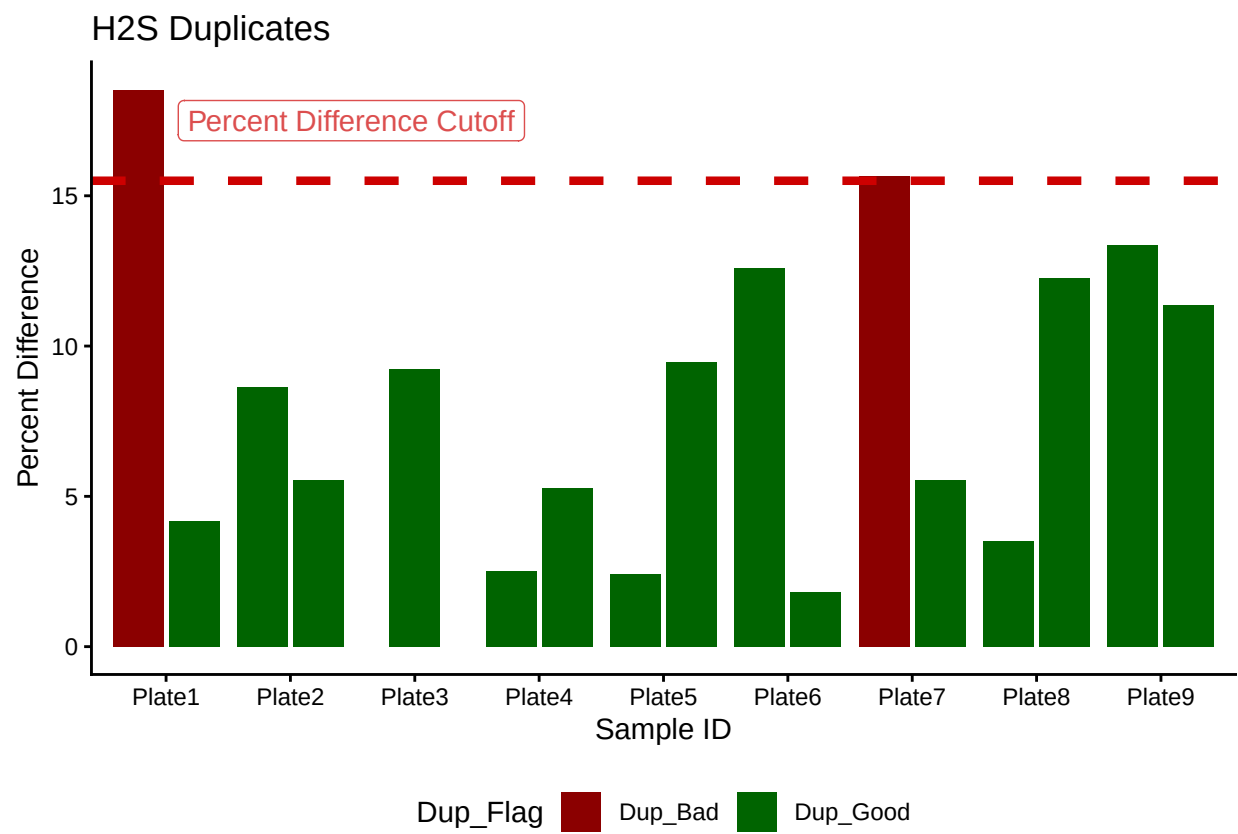
Sample triplicate means and sd dev before bad reps removed



Sample triplicate means and sd dev after bad reps removed



Duplicate QAQC



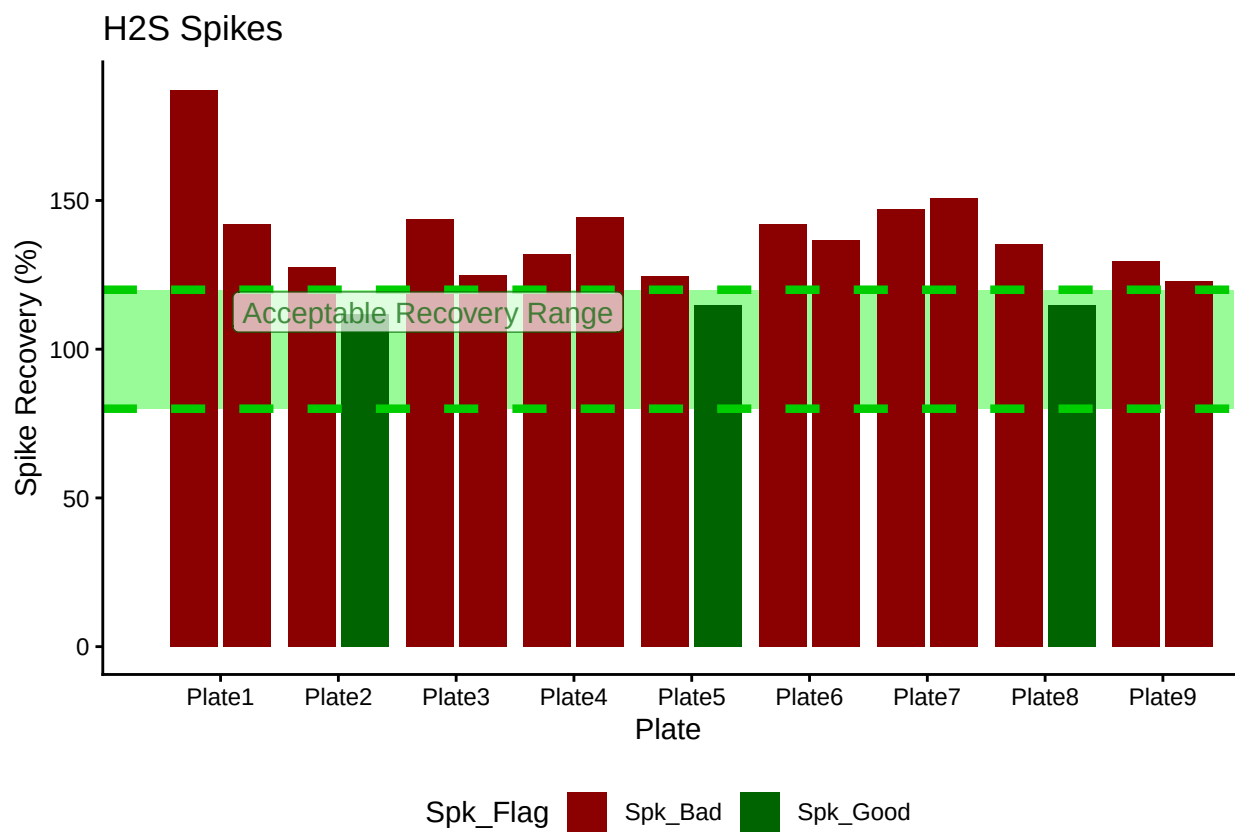
Display Dup Flags

```
## [1] ">60% Dups Pass"
```

| Plate  | Dup_Flag |
|--------|----------|
| Plate1 | Dup_Bad  |
| Plate2 | Dup_Good |
| Plate3 | Dup_Good |
| Plate4 | Dup_Good |
| Plate5 | Dup_Good |
| Plate6 | Dup_Good |
| Plate7 | Dup_Bad  |
| Plate8 | Dup_Good |
| Plate9 | Dup_Good |



Spike QAQC



Display Spike Flags

```
## [1] "Spikes Paired"
```

| Plate  | Spike_Flag |
|--------|------------|
| Plate1 | Spk_Bad    |
| Plate2 | Spk_Bad    |
| Plate3 | Spk_Bad    |
| Plate4 | Spk_Bad    |
| Plate5 | Spk_Bad    |
| Plate6 | Spk_Bad    |
| Plate7 | Spk_Bad    |
| Plate8 | Spk_Bad    |
| Plate9 | Spk_Bad    |

```
## [1] "<60% of Spikes Pass, Rerun"
```

Merge samples with Metadata & check all samples are present

```
## All sample IDs are present in data
```

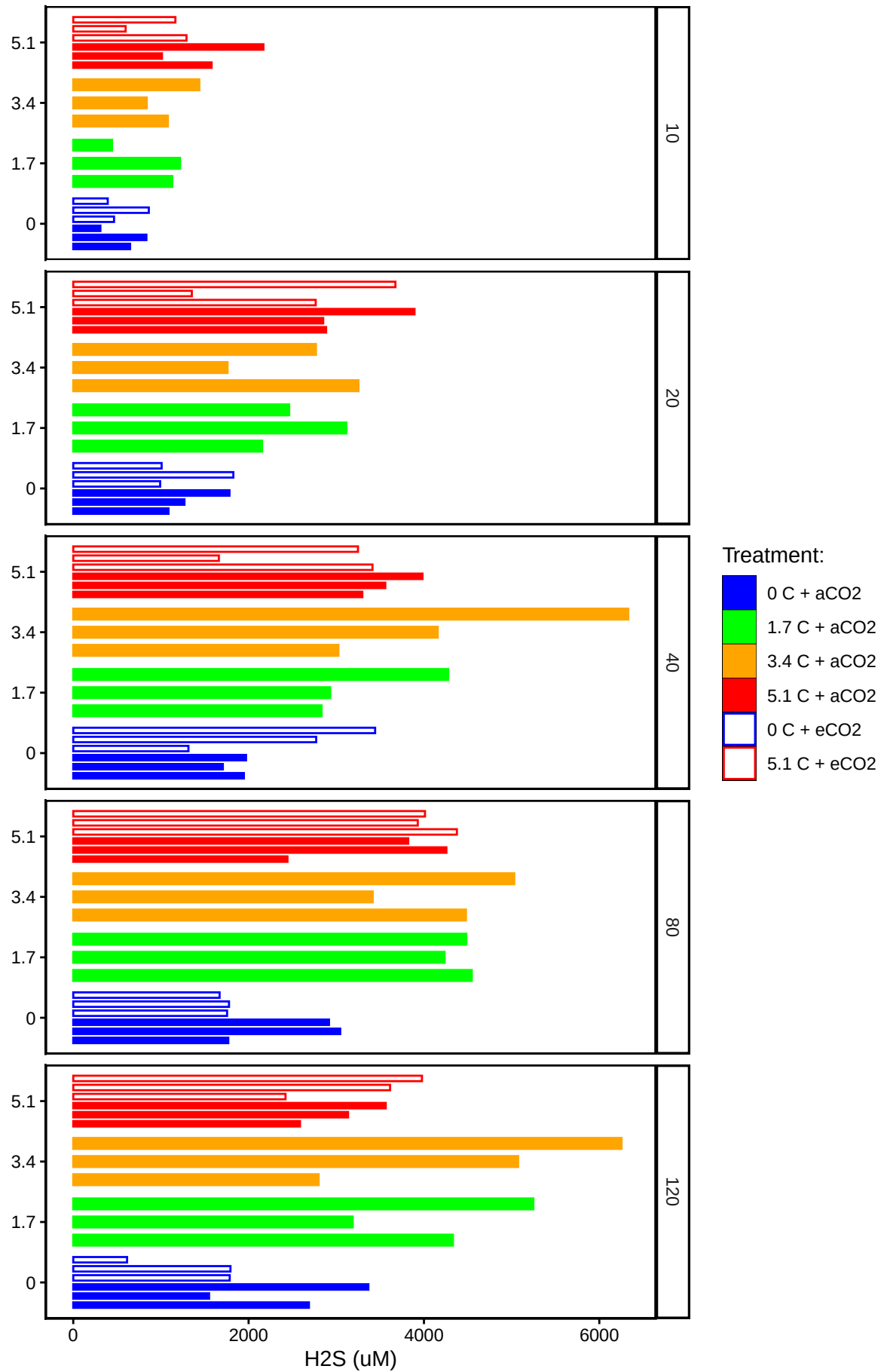
```
##Add Flags
```

```
##Organize Data
```

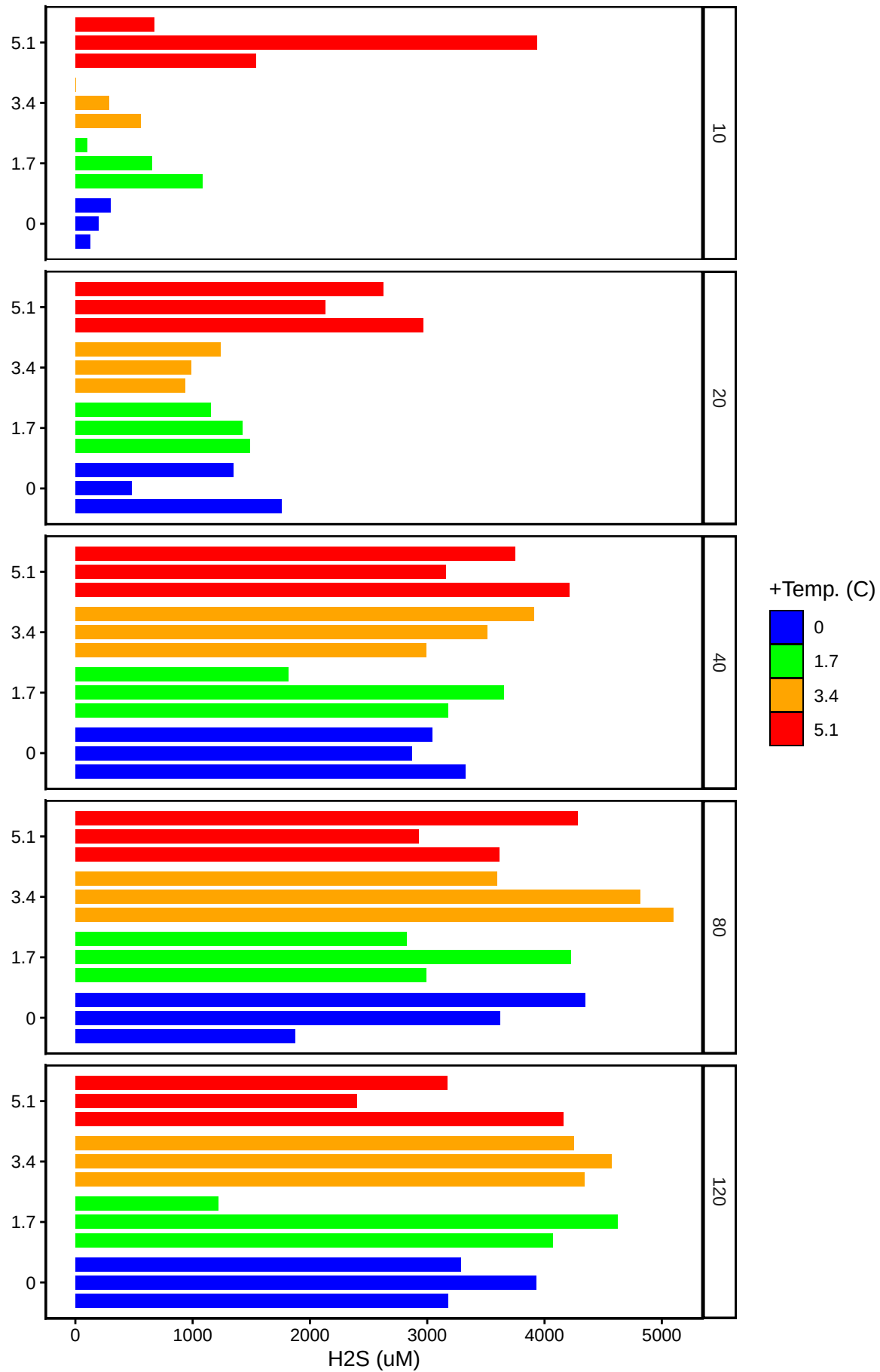
## Visualize Data

```
## Visualize Data
```

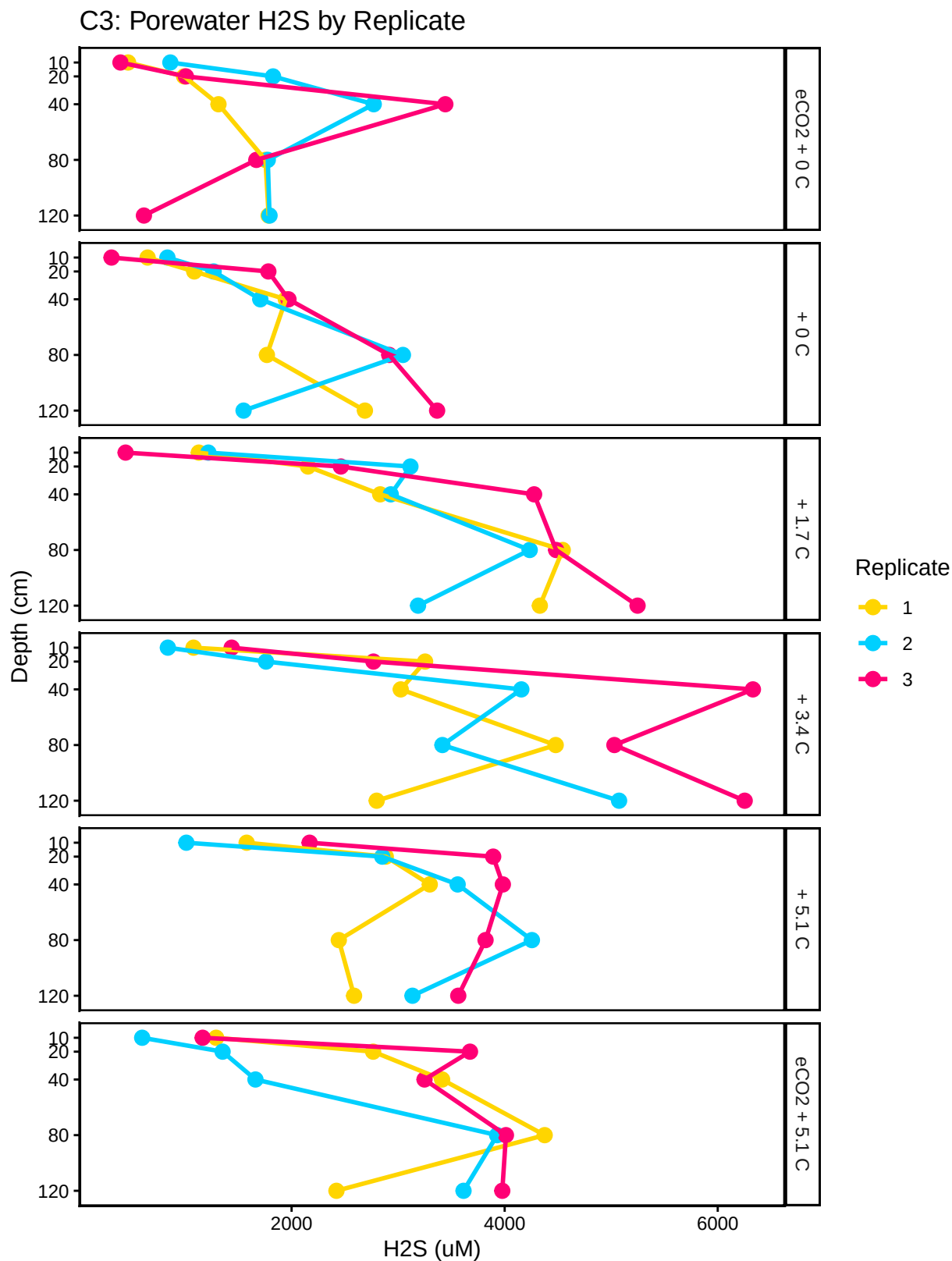
### C3: Porewater H<sub>2</sub>S



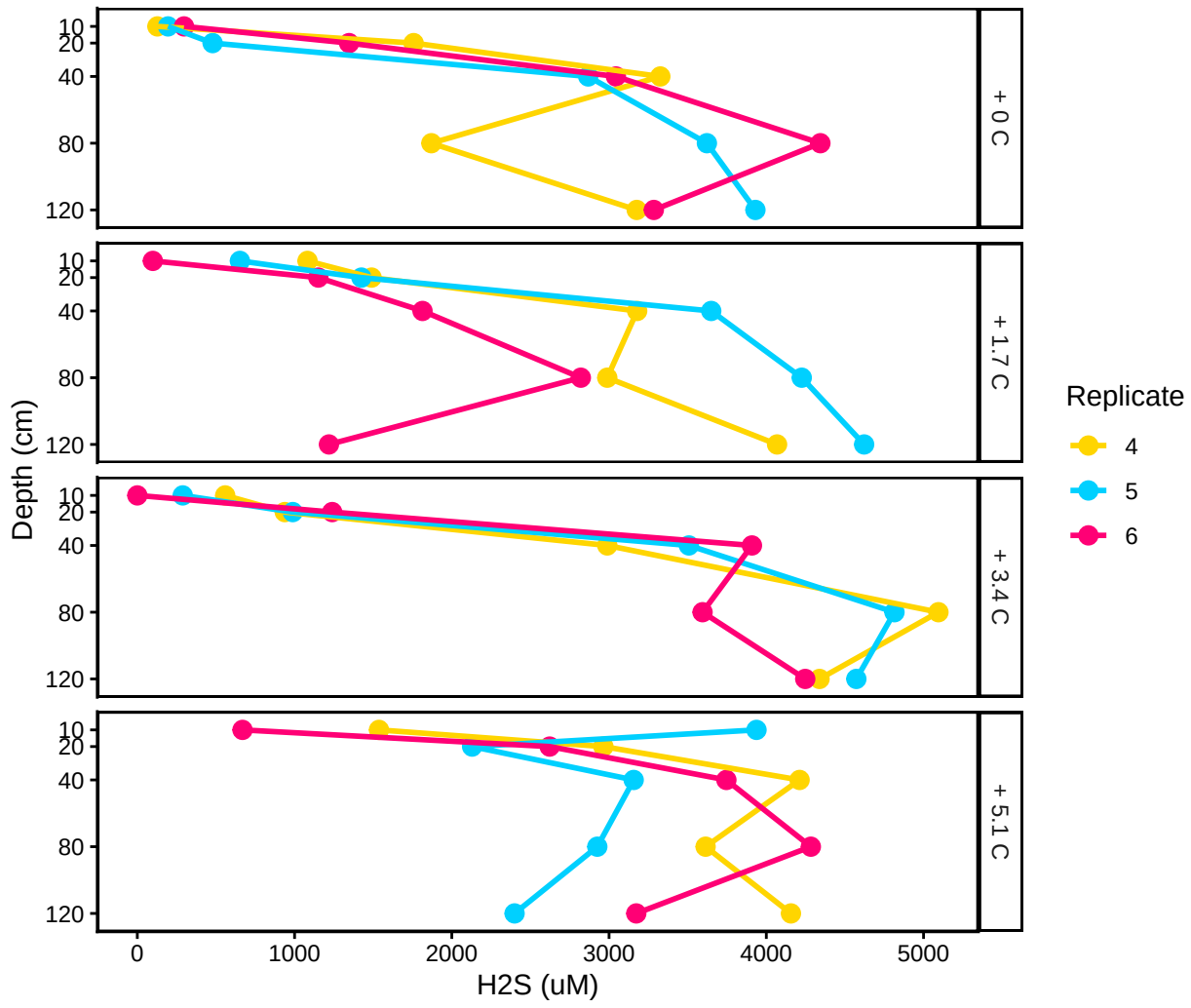
# C4: Porewater H2S



```
## Warning: Using 'size' aesthetic for lines was deprecated in ggplot2 3.4.0.
## i Please use 'linewidth' instead.
## This warning is displayed once per session.
## Call 'lifecycle::last_lifecycle_warnings()' to see where this warning was
## generated.
```



C4: Porewater H2S by Replicate



##Export Data

END